

Auditing Model Substitution in LLM APIs:

Why Software Tests Fail and Hardware Attestation Wins

Are you getting what you pay for? The trust gap in commercial LLM APIs and the case for verifiable inference.

Based on arxiv.org/abs/2504.04715

The Model Substitution Problem: Economic Incentives Drive Covert

Swaps

Commercial LLM API providers charge for access to specific models, but the black-box nature of APIs means users have no guarantee the advertised model is actually being served. Providers face strong economic incentives to covertly substitute cheaper alternatives to reduce GPU costs while maintaining pricing.

Smaller Model

8B parameters served
instead of 70B

Quantized Variant

FP8/INT8 instead
of FP16

Fine-tuned / Updated

Different checkpoint
or revision

Different Family

Entirely different
architecture

Formal Hypothesis Testing Framework

H_0 (Null): The API returns samples from the specified model distribution $P_{\text{spec}}(y|x)$

H_1 (Alternative): The API returns samples from a substituted distribution $P_{\text{actual}}(y|x)$

Given only black-box query access, can an auditor reliably reject H_0 ?

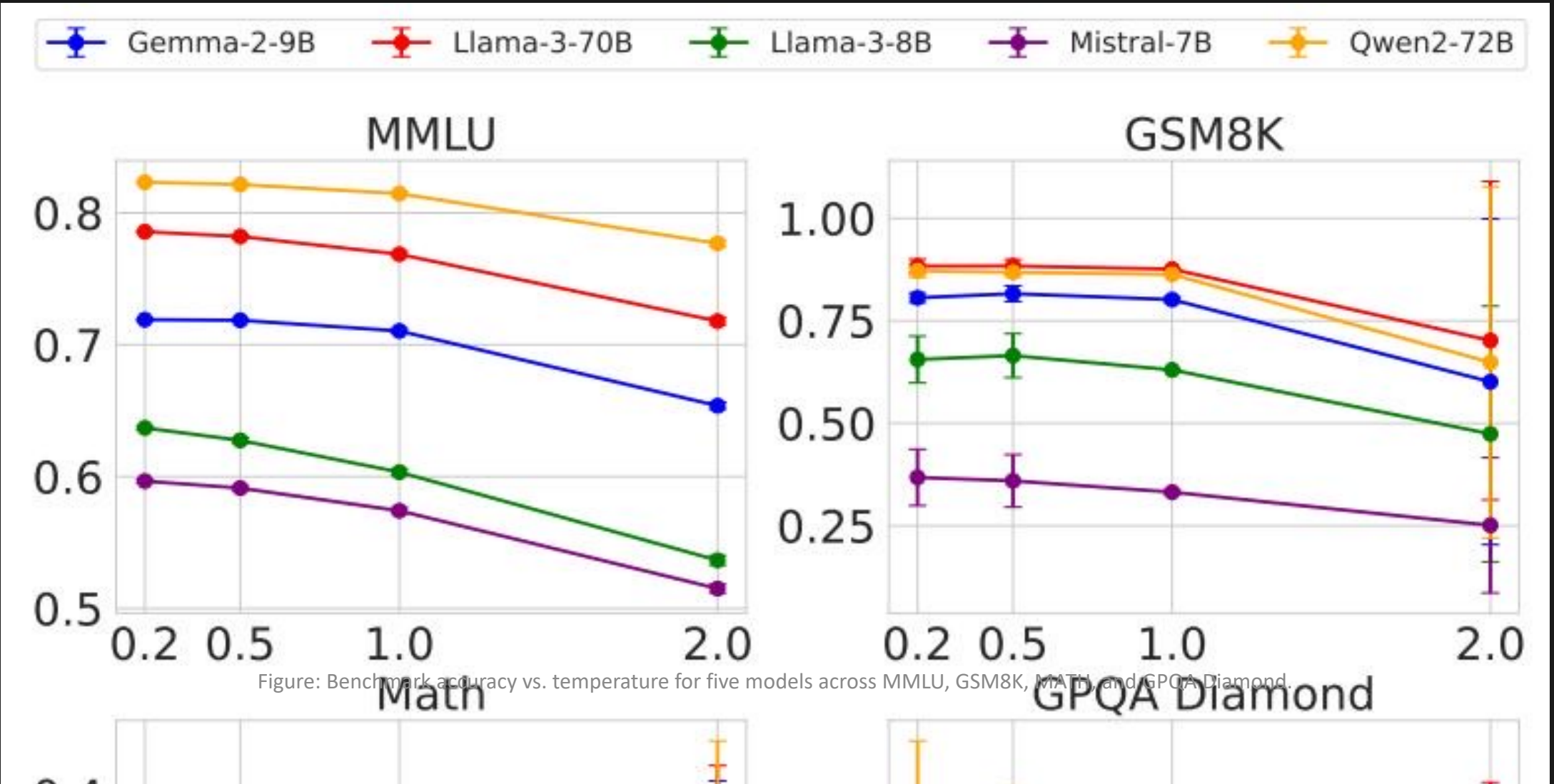
FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Original vs. FP8-quantized scores across four benchmarks for five model families. Drops are often within error bars.

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Higher Temperature Amplifies Performance Gaps but Increases Noise

Benchmark accuracy vs. temperature (0.2, 0.5, 1.0, 2.0) for five models across four tasks. At T=2.0, degradation is universal and the substitution signal is obscured by noise.



Log Probability Fingerprinting Fails Due to Inference Nondeterminism

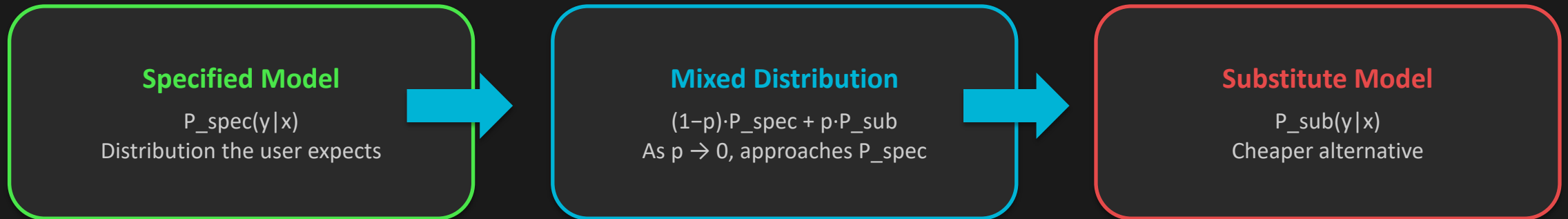
The same Llama-3-8B model produces different token-level log probabilities depending on software framework and GPU hardware.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure: Log-probability variation across inference frameworks (transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2) and hardware (A100 vs H100).

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Providers can route only a fraction p of queries to a cheaper model. As $p \rightarrow 0$, the mixed distribution converges to the specified model, requiring exponentially more queries to detect.



Key Insight

For small p , the mixed distribution is statistically indistinguishable from P_{spec} without an impractically large number of queries. Benchmark evasion (routing known test prompts to the genuine model) further undermines detection.

Adversarial countermeasures make software-only auditing fundamentally query-prohibitive in practice.

Software-Only Auditing Methods Are Fundamentally Unreliable

A structured comparison of three detection paradigms and why each fails under realistic deployment conditions.

Method Type	Core Approach	Weakness	Practical Limitation
Text-based Statistical Tests	Compare output distributions (e.g., MMD) against a local reference	Require enormous query budgets; fail against subtle or randomized substitutions	Query-prohibitive; high false-negative rate
Log-Probability Fingerprinting	Match token-level log-probabilities to a reference model	Production inference is nondeterministic across frameworks and hardware	Cannot distinguish infrastructure differences from actual substitution
Adversarial Countermeasures	Randomized routing + benchmark evasion by the provider	Active evasion makes any software test statistically unreliable	Provider controls routing; auditor has no ground-truth visibility

Conclusion: No software-only method provides reliable, efficient, and robust detection of model substitution in production LLM APIs.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Hardware-secured enclaves shift the trust model from inference monitoring to cryptographic attestation of code and data.

Cryptographic Attestation

Proves specific model weights are loaded and correct inference code is executing inside the enclave.

Modest Overhead

Unlike ZKPs (computationally prohibitive for large models), TEEs incur only modest performance cost.

Practical Deployability

NVIDIA Confidential Computing on H100 GPUs is available today for production workloads.

TEEs vs. Software-Only Methods

- Provable: TEEs — Cryptographic proof of model integrity | Software — Heuristic; no formal guarantee
- Efficient: TEEs — Modest runtime overhead | Software — Query-prohibitive or computationally expensive
- Robust: TEEs — Immune to adversarial routing and evasion | Software — Defeated by randomized substitution and benchmark evasion

Key Takeaways: Close the Verification Gap with Hardware Attestation

1. Economically Motivated Threat

Model substitution is a real, rational strategy for API providers to cut GPU costs while maintaining pricing.

2. FP8 Quantization Is Nearly Invisible

Benchmark drops are small and often within error bars, making statistical detection from accuracy alone infeasible.

3. Inference Nondeterminism Kills Fingerprinting

Log-probability methods fail because the same model produces different metadata across frameworks and hardware.

4. Randomized Substitution Defeats Statistics

Low-rate mixed routing converges to the specified distribution, requiring exponentially more queries.

5. TEEs Are the Only Viable Path

Trusted Execution Environments provide provable, efficient, and robust model integrity — deployable today on NVIDIA H100.

Call to Action: The AI security and cloud infrastructure communities should prioritize TEE-based verifiable inference as a standard requirement for commercial LLM APIs.